

Forecasting with ARIMA models

FISH 550 – Applied Time Series Analysis Download [Rmd pdf](#)

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Forecasting with an ARIMA model

The basic idea of forecasting with an ARIMA model to estimate the parameters and forecast forward.

For example, let's say we want to forecast with a ARIMA(2,1,0) model with drift:

$$z_t = \mu + \beta_1 z_{t-1} + \beta_2 z_{t-2} + e_t$$

where $z_t = x_t - x_{t-1}$, the first difference.

`Arima()` would write this model:

$$(z_t - m) = \beta_1 (z_{t-1} - m) + \beta_2 (z_{t-2} - m) + e_t$$

The relationship between μ and m is $\mu = m(1 - \beta_1 - \beta_2)$.

Let's estimate the β 's for this model from the anchovy data.

```
fit <- forecast::Arima(anchovyts, order=c(2,1,0), include.constant=TRUE)
coef(fit)
```

```
##          ar1          ar2          drift
## -0.53850433 -0.44732522  0.05367062
```

```
mu <- coef(fit)[3]*(1-coef(fit)[1]-coef(fit)[2])
mu
```

```
##          drift
## 0.1065807
```

So we will forecast with this model:

$$z_t = 0.1065807 - 0.53850433z_{t-1} - 0.44732522z_{t-2} + e_t$$

To get our forecast for 1990, we do this

$$(x_{90} - x_{89}) = 0.106 - 0.538(x_{89} - x_{88}) - 0.447(x_{88} - x_{87})$$

Thus

$$x_{90} = x_{89} + 0.106 - 0.538(x_{89} - x_{88}) - 0.447(x_{88} - x_{87})$$

Forecasting with `forecast()`

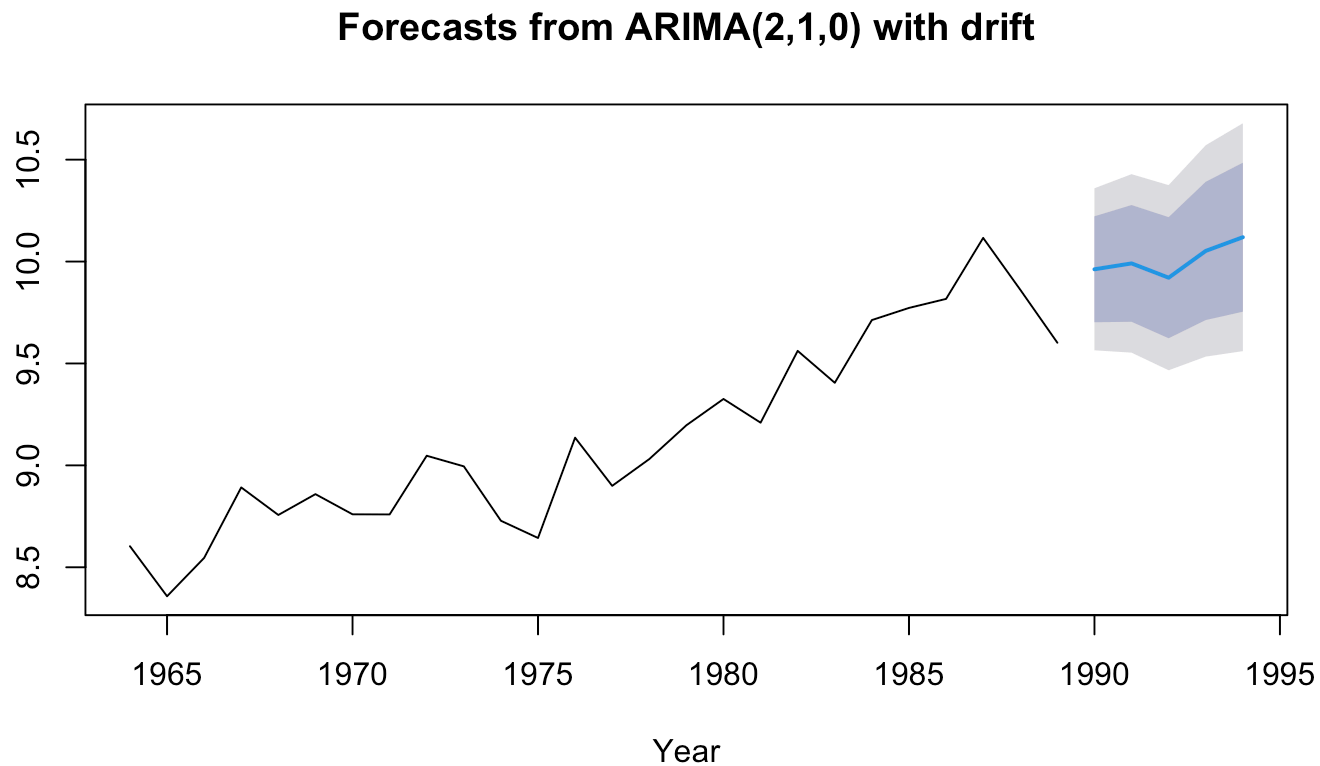
`forecast(fit, h=h)` automates the forecast calculations for us and computes the upper and lower prediction intervals. Prediction intervals include uncertainty in parameter estimates plus the process error uncertainty.

```
fr <- forecast::forecast(fit, h=5)
fr
```

```
##      Point Forecast      Lo 80      Hi 80      Lo 95      Hi 95
## 1990      9.962083  9.702309 10.22186  9.564793 10.35937
## 1991      9.990922  9.704819 10.27703  9.553365 10.42848
## 1992      9.920798  9.623984 10.21761  9.466861 10.37473
## 1993     10.052240  9.713327 10.39115  9.533917 10.57056
## 1994     10.119407  9.754101 10.48471  9.560719 10.67809
```

Plotting our forecasts

```
plot(fr, xlab="Year")
```



Missing values

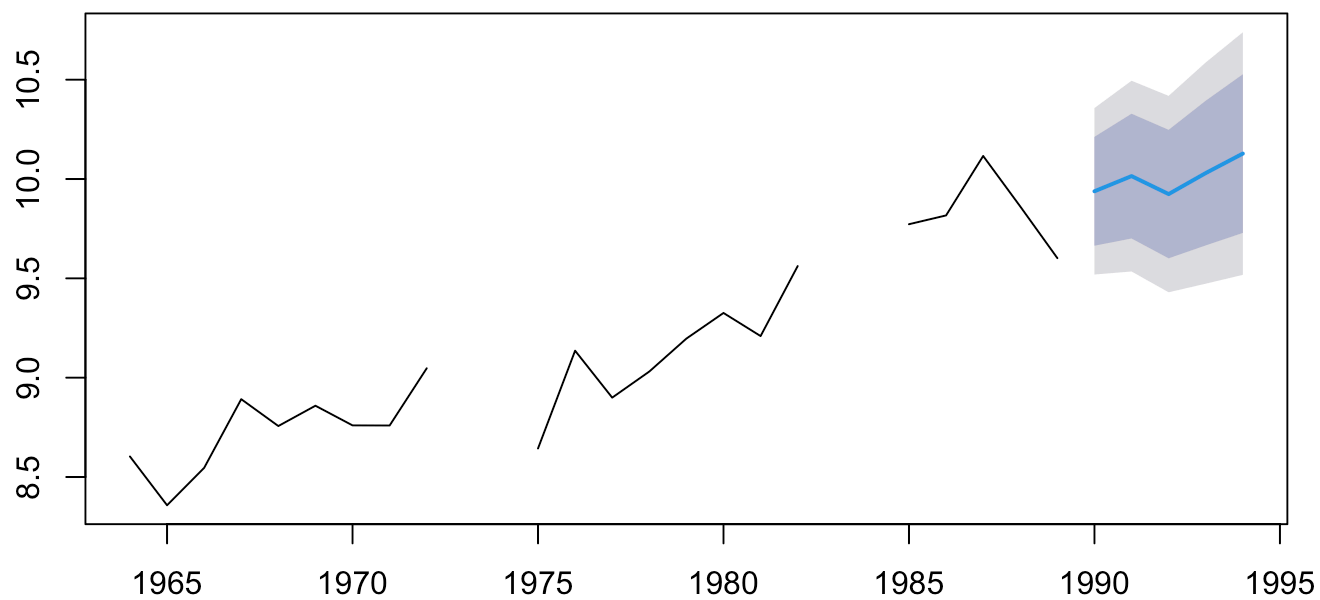
Missing values are allowed for `forecast::Arima()`. We can produce forecasts with the same code.

```
anchovy.miss <- anchovyts
anchovy.miss[10:11] <- NA
anchovy.miss[20:21] <- NA
fit <- forecast::Arima(anchovy.miss, order=c(2,1,0), include.constant=TRUE)
fr <- forecast::forecast(fit, h=5)
fr
```

##	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
## 1990	9.938269	9.664479	10.21206	9.519543	10.35700
## 1991	10.014686	9.700961	10.32841	9.534885	10.49449
## 1992	9.924208	9.601147	10.24727	9.430129	10.41829
## 1993	10.029988	9.666069	10.39391	9.473421	10.58656
## 1994	10.128066	9.729066	10.52707	9.517848	10.73828

```
plot(fr)
```

Forecasts from ARIMA(2,1,0) with drift



Using auto.arima()

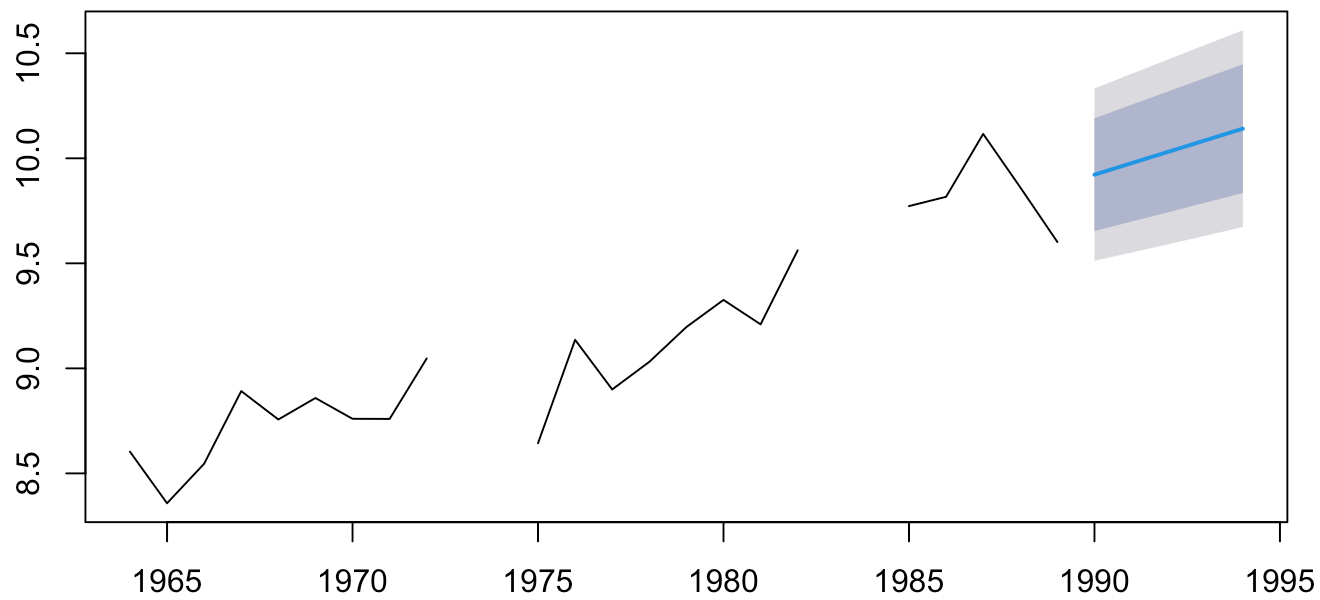
We can let forecast to select the ARIMA model:

```
anchovy.miss <- anchovyts
anchovy.miss[10:11] <- NA
anchovy.miss[20:21] <- NA
fit <- forecast::auto.arima(anchovy.miss)
fit

## Series: anchovy.miss
## ARIMA(0,1,1) with drift
##
## Coefficients:
##           ma1    drift
##      -0.7240  0.0548
## s.e.   0.2283  0.0125
##
## sigma^2 = 0.04355: log likelihood = 3.52
## AIC=-1.03  AICc=0.11  BIC=2.63
```

```
fr <- forecast::forecast(fit, h=5)  
plot(fr)
```

Forecasts from ARIMA(0,1,1) with drift



Seasonal ARIMA Models

Load the chinook salmon data set

```
load("chinook.RData")  
head(chinook)
```

##	Year	Month	Species	log.metric.tons	metric.tons
## 1	1990	Jan	Chinook	3.397858	29.9
## 2	1990	Feb	Chinook	3.808882	45.1
## 3	1990	Mar	Chinook	3.511545	33.5
## 4	1990	Apr	Chinook	4.248495	70.0
## 5	1990	May	Chinook	5.200705	181.4
## 6	1990	Jun	Chinook	4.371976	79.2

The data are monthly and start in January 1990. To make this into a ts object do

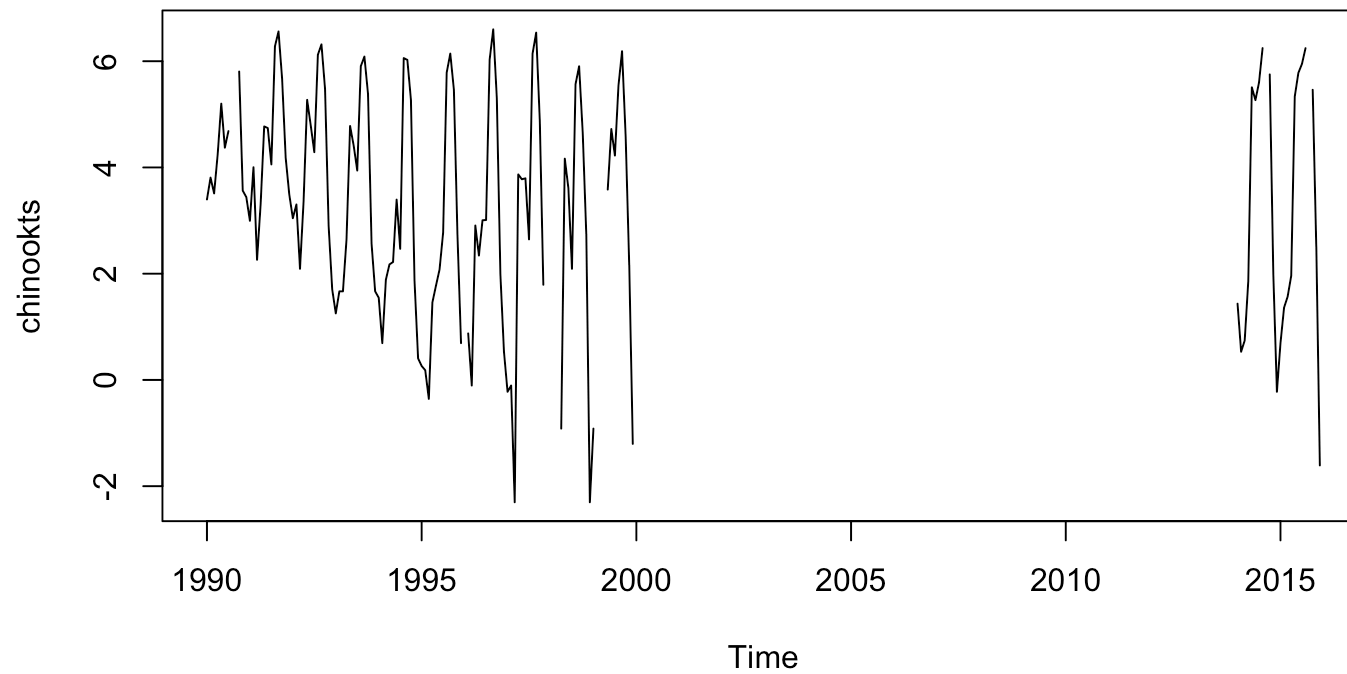
```
chinookts <- ts(chinook$log.metric.tons, start=c(1990,1),  
               frequency=12)
```

start is the year and month and frequency is the number of months in the year.

Use `?ts` to see more examples of how to set up ts objects.

Plot seasonal data

```
plot(chinookts)
```



Seasonal ARIMA model: SARIMA

Basic structure of a seasonal AR model

differentiated data = AR(non-seasonal) + AR(seasonal) + cross-terms + MA

Differenced chinook data

- No differencing x_t . We are modeling the data.
- $x_t - x_{t-1}$. We are modeling the monthly change.
- $x_t - x_{t-12}$. We are modeling the seasonal change. Jan vs Jan last year, Feb vs Feb last year
- $(x_t - x_{t-12}) - (x_{t-1} - x_{t-13})$. We are modeling the seasonal rate of monthly change.
- $(x_t - x_{t-1}) - (x_{t-12} - x_{t-13})$. We are modeling the seasonal monthly change.

Notation

ARIMA (p,d,q)(ps,ds,qs)S

ARIMA (non-seasonal part)(seasonal part)Frequency

ARIMA (non-seasonal) means t correlated with $t - 1$

ARIMA (seasonal) means t correlated with $t - s$

Seasonal autoregressive

ARIMA (1,0,0)(1,0,0)[12]

This month depends on last month plus the same month last year

What data are we modeling? Zero differences, so

$$z_t = x_t$$

Write out the AR parts with z_t

$$z_t = \phi_1 z_{t-1} + \Phi_1 z_{t-12} - \text{cross-term} + w_t$$

No MA in this model so $w_t = e_t$ is white noise.

Autocorrelated seasonal differences

ARIMA (1,0,0)(1,1,0)[12]

The seasonal difference this month depends on the seasonal difference last month plus the seasonal difference last year

z_t is just a seasonal difference.

$$z_t = x_t - x_{t-12}$$

Write out the AR parts with z_t

$$z_t = \phi_1 z_{t-1} + \Phi_1 z_{t-12} - \text{cross-term} + w_t$$

No MA in this model so $w_t = e_t$ is white noise.

Seasonal random walk model

ARIMA(0,0,0)(0,1,0)[12]

This month's value is the same as last year's, plus some random noise

expected January 1990 = January 1989 + constant mean

$$Z_t = x_t - x_{t-12}$$

No AR nor MA

$$Z_t = w_t = x_t - x_{t-12}$$

$$x_t = x_{t-12} + w_t$$

Seasonal random walk model with random trend

ARIMA(0,1,0)(0,1,0)[12]

expected Feb 1990 = Feb 1989 + (Jan 1990 - Jan 1989)

$$z_t = (x_t - x_{t-12}) - (x_{t-1} - x_{t-13})$$

No AR or MA part.

$$z_t = w_t = (x_t - x_{t-12}) - (x_{t-1} - x_{t-13})$$

$$x_t = x_{t-12} + (x_{t-1} - x_{t-13}) + w_t$$

airline model

ARIMA(0, 1, 1)(0, 1, 1)[12]

Seasonal random walk model with random trend + autocorrelated noise

$$z_t = (x_t - x_{t-12}) - (x_{t-1} - x_{t-13})$$

No AR part. Write out the MA parts, the w_t .

$$w_t = e_t - \theta_1 e_{t-1} - \Theta_1 e_{t-12} + \text{cross-term}$$

auto.arima() for seasonal ts

`auto.arima()` will recognize that our data has season and fit a seasonal ARIMA model to our data by default. We will define the training data up to 1998 and use 1999 as the test data.

```
traindat <- window(chinookts, c(1990,10), c(1998,12))
testdat <- window(chinookts, c(1999,1), c(1999,12))
fit <- forecast::auto.arima(traindat)
fit
```

```
## Series: traindat
## ARIMA(1,0,0)(0,1,0)[12] with drift
##
## Coefficients:
##          ar1      drift
##      0.3676  -0.0320
## s.e.  0.1335   0.0127
##
## sigma^2 = 0.8053:  log likelihood = -107.37
## AIC=220.73   AICc=221.02   BIC=228.13
```

Forecasting with a Seasonal model

Load the chinook salmon data

```
library(atsalibrary)
```

```
## Warning: package 'atsalibrary' was built under R version  
## 4.4.3
```

```
## Loading required package: broom
```

```
## Loading required package: corrplot
```

```
## corrplot 0.95 loaded
```

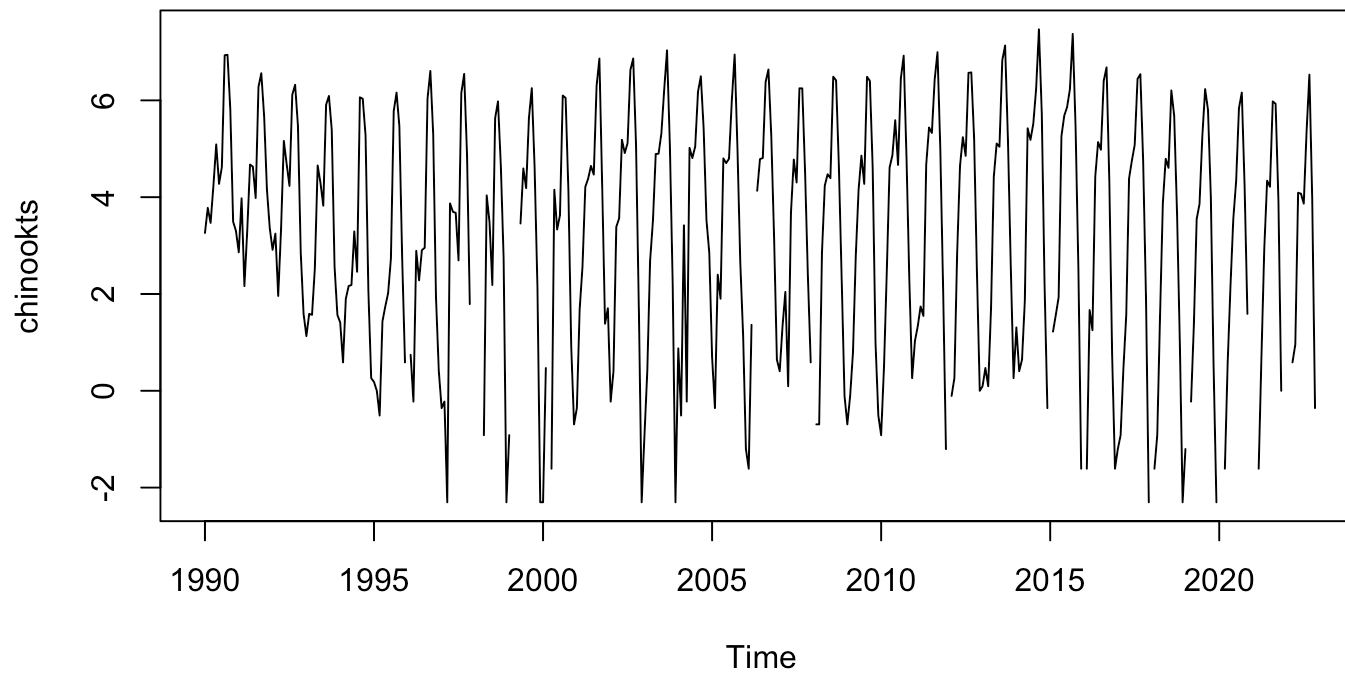
```
## Loading required package: dplyr
```

```
##  
## Attaching package: 'dplyr'
```

```
## The following object is masked from 'package:gridExtra':  
##  
##
```

Plot seasonal data

```
plot(chinookts)
```



Fit

Missing values are ok when fitting a seasonal ARIMA model

```
traindat <- window(chinookts, c(1990,10), c(1998,12))  
testdat <- window(chinookts, c(1999,1), c(1999,12))  
fit <- forecast::auto.arima(traindat)  
fit
```

```
## Series: traindat  
## ARIMA(1,0,0)(0,1,0)[12] with drift  
##  
## Coefficients:  
##          ar1    drift  
##      0.3728 -0.0315  
## s.e.  0.1332  0.0127  
##  
## sigma^2 = 0.7918:  log likelihood = -106.67  
## AIC=219.35   AICc=219.64   BIC=226.75
```

Forecast using seasonal model

```
fr <- forecast::forecast(fit, h=12)  
plot(fr)  
points(testdat)
```

Forecasts from ARIMA(1,0,0)(0,1,0)[12] with drift

